

Approximating Undulator Power with the Bending-Magnet Formula

Theory, Validity, and Practical Implementation

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1 Introduction

Undulators produce quasi-monochromatic, highly collimated radiation through the constructive interference of emission from successive magnetic periods. Computing the exact spectrum requires evaluating the full radiation integral (e.g. via SRW or SPECTRA), which is numerically expensive and requires detailed knowledge of the electron beam emittance.

For *power* estimates—particularly to assess thermal load on optical elements—the bending-magnet (BM) approximation provides a fast, analytical alternative. The key insight is that in the wiggler limit ($K \gg 1$) successive poles radiate *incoherently*, each like an independent bending magnet. The accuracy of this approximation and its practical implementation are discussed in Section 2.

2 Theoretical Motivation and Implementation

The deflection parameter K characterises the amplitude of the electron’s transverse oscillation in the periodic field $B_y = B_0 \cos(2\pi z/\lambda_u)$:

$$K = \frac{eB_0\lambda_u}{2\pi m_e c} \approx 0.9337 B_0 [\text{T}] \lambda_u [\text{cm}]. \quad (1)$$

The peak angular excursion of the electron is $\theta_{\text{max}} = K/\gamma$, where $\gamma = E_e/m_e c^2$ is the Lorentz factor.

Two radiation regimes can be considered:

- **Undulator regime** ($K \lesssim 1$): $\theta_{\max} \lesssim 1/\gamma$. Emission cones from adjacent poles overlap; interference produces sharp harmonic peaks. On-axis brightness scales as N^2 .
- **Wiggler regime** ($K \gg 1$): $\theta_{\max} \gg 1/\gamma$. Emission fans are well separated; poles radiate independently. The spectrum approaches $2N$ times a bending-magnet spectrum.

2.1 Power of the full emission of an insertion device

The total radiated power of any insertion device with sinusoidal field $B(z) = B_0 \cos(2\pi z/\lambda_u)$ is [4]:

$$P [\text{W}] = 633 B_0^2 [\text{T}^2] L [\text{m}] E_e^2 [\text{GeV}^2] I [\text{A}] \quad (2)$$

where $L = N\lambda_u$ is the total device length. The coefficient 633 incorporates the sinusoidal average $\langle B^2 \rangle = B_0^2/2$. This formula is *exact for any* K ; only the spectral *distribution* depends on the regime. In terms of K :

$$P [\text{kW}] = 7.257 \times 10^{-5} \frac{E_e^2 [\text{GeV}^2] K^2 N}{\lambda_u [\text{m}]} I [\text{A}]. \quad (3)$$

2.2 Power on a slit of an insertion device

While the total power, Eq. (2), fixes the overall scale, sizing a slit and estimating the power it intercepts require the *angular* distribution of the radiated power. Following Kim [4], the frequency-integrated power density factorises into its on-axis value and a normalised angular shape function (Kim Eq. 4.61),

$$\frac{d^2 P}{d\Omega}(\theta, \psi) = \frac{d^2 P}{d\Omega} \Big|_0 f_K(\gamma\theta, \gamma\psi), \quad (4)$$

where θ and ψ are the horizontal and vertical observation angles. The on-axis density, in practical units (Kim Eq. 4.63), is

$$\frac{d^2 P}{d\Omega} \Big|_0 = 10.84 B_0 [\text{T}] E_e^4 [\text{GeV}^4] I [\text{A}] N G(K) \quad [\text{W/mrad}^2], \quad (5)$$

where the $2N$ factor for the $2N$ poles and the interference factor $G(K)$ are both absorbed in the coefficient ($10.84 = 2 \times 5.42$, with N written explicitly). The interference factor (Kim Eq. 4.60) is

$$G(K) = \frac{K(K^6 + \frac{24}{7}K^4 + 4K^2 + \frac{16}{7})}{(1 + K^2)^{7/2}}, \quad G(K) \rightarrow 1 \text{ for } K \gtrsim 1. \quad (6)$$

The normalised angular function is valid for any K , satisfies $f_K(0, 0) = 1$, and reads (Kim Eq. 4.62)

$$f_K(\gamma\theta, \gamma\psi) = \frac{16K}{7\pi G(K)} \int_{-\pi}^{\pi} \frac{\sin^2 \xi}{D^5} \left[(1 + X^2 - Y^2)^2 + 4X^2 Y^2 \right] d\xi, \quad (7)$$

with

$$X = \gamma\psi, \quad Y = K \cos \xi - \gamma\theta, \quad D = 1 + X^2 + Y^2. \quad (8)$$

The power transmitted by a rectangular slit subtending horizontal half-angle θ_0 and vertical half-angle ψ_0 is then the angular density integrated over the slit acceptance,

$$P_{\text{slit}} = \frac{d^2 P}{d\Omega} \Big|_0 \int_{-\theta_0}^{\theta_0} \int_{-\psi_0}^{\psi_0} f_K(\gamma\theta, \gamma\psi) d\theta d\psi \quad (9)$$

with the angles expressed in mrad to match the units of Eq. (5). Integrating over all angles recovers the total power, Eq. (2). Equations (4)–(9) are exact within the single-electron, zero-emittance far-field approximation; the bending-magnet treatment developed below replaces the exact f_K by a cheaper incoherent-pole estimate suited to fast thermal-load calculations.

2.3 On-Axis Brilliance: Wiggler versus Undulator

We have seen that the expressions for the total power emitted by the undulator and wiggler are the same, when integrated over all the photon energies. The headline distinction lies in how the radiation from successive magnet periods combines, and it appears directly in the scaling with the number of periods N . The two regimes are set by the deflection parameter K which compares the electron's angular excursion $\sim K/\gamma$ with the natural radiation opening angle $1/\gamma$.

When $K \lesssim 1$ (undulator regime), the angular excursion is comparable to $1/\gamma$, so the radiation cones from all N periods overlap and interfere. On resonance the field *amplitudes* add, and the on-axis angular flux density scales as N^2 :

$$\left. \frac{dF}{d\Omega} \right|_0 \approx 1.744 \times 10^{14} N^2 E^2 [\text{GeV}] I [\text{A}] F_n(K) \quad [\text{ph/s/mrad}^2/0.1\% \text{ BW}]. \quad (10)$$

The interference also squeezes the radiation into a narrow central cone of divergence

$$\sigma'_r \approx \sqrt{\frac{\lambda}{2L}} \sim \frac{1}{\gamma\sqrt{N}}, \quad L = N\lambda_u, \quad (11)$$

and concentrates the spectrum into sharp harmonic lines of bandwidth $\Delta\lambda/\lambda \sim 1/(nN)$. Since brilliance divides the flux by the source phase space,

$$B \approx \frac{F}{4\pi^2 \sigma_x \sigma'_x \sigma_y \sigma'_y}, \quad (12)$$

the small divergence and spectral concentration raise it further. The undulator on-axis brilliance therefore scales roughly as N^2 , with additional gains from collimation and monochromaticity.

When $K \gg 1$ (wiggler regime), the angular swing K/γ greatly exceeds $1/\gamma$, so the cones from different poles point in different directions and the $2N$ poles behave like independent bending magnets. The *intensities* add, giving a linear scaling:

$$\left. \frac{dF}{d\Omega} \right|_0 \approx 1.327 \times 10^{13} (2N) E^2 [\text{GeV}] I [\text{A}] H_2(y) \quad [\text{ph/s/mrad}^2/0.1\% \text{ BW}]. \quad (13)$$

The radiation fans out horizontally over $\pm K/\gamma$ and forms a broadband continuum (as for a bending magnet, but with higher critical energy). Both effects enlarge the phase space and dilute the spectral density, so the brilliance remains linear in N with no interference enhancement.

For the same N , the undulator on-axis brilliance exceeds the wiggler's by roughly a factor of N (the ratio N^2/N), before even accounting for the undulator's smaller divergence and narrower bandwidth—so in practice the gap is often several orders of magnitude. The trade-off is that wigglers deliver higher integrated flux and reach harder X-ray energies across a broad continuum, whereas undulators dominate in brilliance, coherence, and tunable monochromatic lines. Physically, the crossover is simply whether K/γ is smaller than (undulator) or much larger than (wiggler) the $1/\gamma$ radiation cone.

2.4 The undulator spectrum approximated by the wiggler

In the wiggler limit, the undulator spectrum will show a multitude of peaks that tend to merge and approaches $2N$ times the spectrum of an "averaged" single bending magnet. Figure 1 compares the spectral power computed by exact (SRW in OASYS/XOPPY Undulator Spectrum), wiggler (WS in OASYS/XOPPY WS). Two cases are compared, the full emission (or radiation collected in a very wide slit, in this case $30 \times 10 \text{ mm}^2$ at 23 m) or in a "small slit" as defined by the front end and primary slit (FE aperture is $1.8 \times 1.0 \text{ mm}^2$ at 23 m). The cumulated power plots show that

the total integrated power is the same full emission and emission on a small aperture (equations 2 and 9, respectively). The cumulated power vs photon energy shows steps for the undulator and is very smooth for the wiggler, meaning that if some non-constant filtering is applied (when using attenuators and mirrors) the total integrated values will show differences. However, in cases where the filtering is not very selective, i.e., it is applied over undulator peaks that average out on the wiggler continuum, which appears for K larger than one and is the case when the total power is higher, the use of a wiggler approximation may be reasonable for calculating the total power. This is exploited in this document.

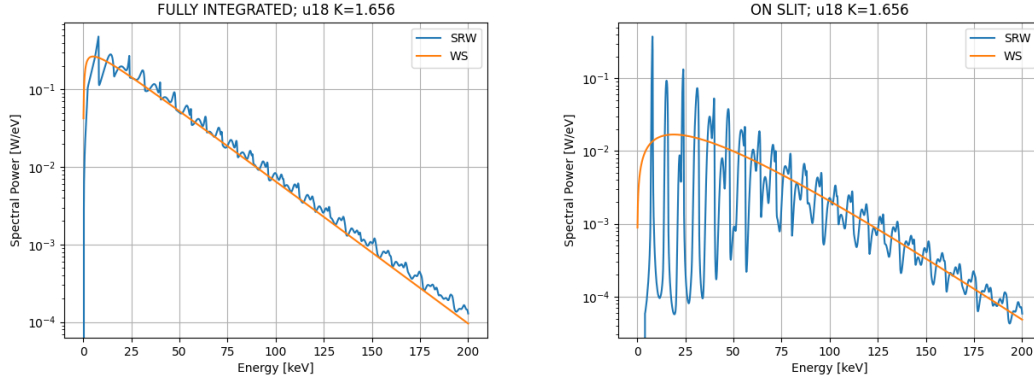


Figure 1: Spectral power for a large slit (left, $30 \times 10 \text{ mm}^2$) and a small slit (right, $1.8 \times 1.0 \text{ mm}^2$) at 23 m.

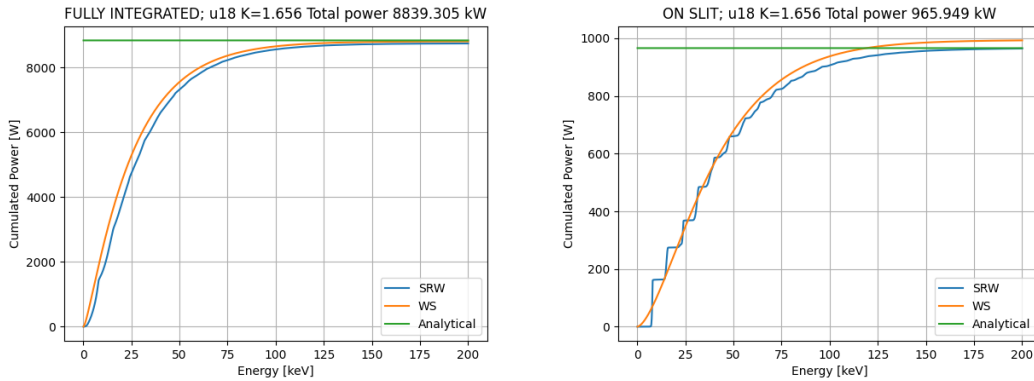


Figure 2: Cumulated power for a large slit (left, $30 \times 10 \text{ mm}^2$) and a small slit (right, $1.8 \times 1.0 \text{ mm}^2$) at 23 m. (For the small slit WS slightly overestimates the integrated power.)

2.5 Bending magnet radiation

The angle-integrated BM spectral flux [ph/s/0.1%bw/mrad] is [2]:

$$\Phi_{\text{BM}}(E, E_c) = 2.45706 \times 10^{13} E_e [\text{GeV}] I [\text{A}] G_1(E/E_c), \quad (14)$$

where $G_1(y) = y \int_y^\infty K_{5/3}(x) dx$ is a universal synchrotron function and the critical energy is

$$E_c [\text{keV}] = 0.6650 E_e^2 [\text{GeV}^2] B [\text{T}]. \quad (15)$$

The exact integral of G_1 is obtained by exchanging the order of integration:

$$\int_0^\infty G_1(y) dy = \int_0^\infty K_{5/3}(x) \frac{x^2}{2} dx = \Gamma\left(\frac{7}{3}\right) \Gamma\left(\frac{2}{3}\right) = \frac{8\pi}{9\sqrt{3}} \approx 1.6124. \quad (16)$$

The power per unit horizontal angle is obtained by integrating Eq. (14) over photon energy, using $dP/dE = \Phi_{\text{BM}}(E) \times E[\text{eV}] \times e/(0.001 \times E[\text{eV}])$:

$$\frac{dP}{d\theta} [\text{W/mrad}] = 2.45706 \times 10^{13} \times 1.602 \times 10^{-16} \times \frac{8\pi}{9\sqrt{3}} \times 665 E_e^3 [\text{GeV}] B [\text{T}] I [\text{A}] = 4.221 E_e^3 B I. \quad (17)$$

For a constant-field BM of arc length L , the total bending angle is $\Delta\theta = L/\rho$ with $B\rho [\text{T m}] = 3.3356 E_e [\text{GeV}]$, so integrating Eq. (17) over the arc gives:

$$P_{\text{BM}} = \frac{4.221 \times 10^3}{3.3356} E_e^2 B^2 L I \approx 1268 E_e^2 [\text{GeV}^2] B^2 [\text{T}^2] L [\text{m}] I [\text{A}] \quad [\text{W}]. \quad (18)$$

Independent derivation from the relativistic Larmor formula

The same result can be derived from first principles using the relativistic Larmor formula for a charge undergoing circular motion [1] (eq. 14.31, SI units):

$$P_e = \frac{e^2 c}{6\pi\epsilon_0} \cdot \frac{\gamma^4}{R^2} \quad [\text{W/electron}]. \quad (19)$$

For a beam of current I , the number of electrons simultaneously present in an arc of length $L = R\Delta\theta$ is

$$\bar{n} = \frac{I}{e} \cdot \frac{R\Delta\theta}{c}, \quad (20)$$

so the total beam power radiated in that arc is

$$P = P_e \cdot \bar{n} = \frac{e^2 c}{6\pi\epsilon_0} \cdot \frac{\gamma^4}{R^2} \cdot \frac{I R \Delta\theta}{e c} = \frac{e}{6\pi\epsilon_0} \cdot \frac{\gamma^4 \Delta\theta I}{R}. \quad (21)$$

Substituting $R[\text{m}] = 3.3356 E_e [\text{GeV}]/B[\text{T}]$ and $\gamma = 1956.95 E_e [\text{GeV}]$:

$$P = \frac{e (1956.95)^4}{6\pi\epsilon_0 \times 3.3356} \cdot E_e^3 B \Delta\theta I. \quad (22)$$

Evaluating the prefactor with $e = 1.602 \times 10^{-19} \text{ C}$, $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$:

$$\frac{e (1956.95)^4}{6\pi\epsilon_0 \times 3.3356} = \frac{1.602 \times 10^{-19} \times 1.463 \times 10^{13}}{6\pi \times 8.854 \times 10^{-12} \times 3.3356} = \frac{2.344 \times 10^{-6}}{5.567 \times 10^{-10}} = 4211 \quad [\text{W rad}^{-1} \text{GeV}^{-3} \text{T}^{-1} \text{A}^{-1}], \quad (23)$$

giving

$$\boxed{\frac{dP}{d\theta} = 4211 \cdot E_e^3 [\text{GeV}^3] \cdot B [\text{T}] \cdot I [\text{A}] \quad [\text{W/rad}]} = 4.211 \cdot E_e^3 \cdot B \cdot I \quad [\text{W/mrad}]. \quad (24)$$

This agrees with the spectral-formula result of 4.221 W/mrad (Eq. 17) to within **0.4%**, the residual being purely rounding of the fundamental constants. The two routes — relativistic Larmor from first principles, and integration of the BM spectral formula with $\int_0^\infty G_1 dy = 8\pi/(9\sqrt{3})$ — are therefore fully consistent.

2.6 The wiggler approximated by the BM

In the wiggler limit the $2N$ half-periods each contribute an independent BM arc. Two complementary approximations are derived below, valid in different aperture regimes.

Approach 1 — arc integration (full aperture)

The total power is obtained by applying Eq. (17) locally along the sinusoidal arc $B(s) = B_0 \cos(2\pi s/\lambda_u)$, with $d\theta = |B(s)|/(B\rho) ds$:

$$\begin{aligned} P &= 2N \int_0^{\lambda_u/2} \frac{dP}{d\theta} \Big|_{B(s)} \frac{|B(s)|}{B\rho} ds \\ &= \frac{2N \times 4.229 \times 10^3}{3.3356} E_e^2 I \int_0^{\lambda_u/2} B_0^2 \cos^2\left(\frac{2\pi s}{\lambda_u}\right) ds \\ &= \frac{2N \times 4.229 \times 10^3}{3.3356} E_e^2 I \cdot \frac{B_0^2 \lambda_u}{4}. \end{aligned} \quad (25)$$

This gives coefficient $634 \text{ W T}^{-2} \text{ m}^{-1} \text{ GeV}^{-2} \text{ A}^{-1}$, in excellent agreement with the LBL reference value [4]:

$$P [\text{W}] = 633 B_0^2 [\text{T}^2] L [\text{m}] E_e^2 [\text{GeV}^2] I [\text{A}], \quad (26)$$

where $L = N\lambda_u$. The factor $\langle B^2 \rangle = B_0^2/2$ from the sinusoidal average enters naturally through the arc integral, recovering Eqs. (2) and (3) stated above. In practice Approach 1 is computed numerically: N_{arc} BM spectra are evaluated at local $E_c(s) = 665 E_e^2 |B(s)|$ and summed weighted by local $d\theta(s)$. This yields a softer spectrum than the single peak- B_0 call of Approach 2, in better agreement with WS for large apertures.

Limitation: Approach 1 integrates over all arc points regardless of horizontal angle; it cannot account for a finite slit that selects only part of the fan.

Application: Approach 1 is used in OASYS/XOPPY Wiggler (full emission) and in shadow4.

Approach 2 — peak field with slit acceptance

When a slit of width $w_h \times w_v$ at distance D clips the horizontal fan, the angular acceptances are $\Delta\theta_h^{\text{slit}} = w_h/D$ and $\Delta\psi^{\text{slit}} = w_v/D$. A narrow slit preferentially selects radiation from the *turning points* of the electron trajectory where $|B| \approx B_0$, so the BM spectrum at peak critical energy $E_c(B_0)$ is appropriate. The effective horizontal divergence is

$$\Delta\theta_h^{\text{used}} = \min\left(\frac{2K}{\gamma}, \frac{w_h}{D}\right), \quad (27)$$

and the slit-integrated flux is

$$\Phi_{\text{slit}}(E) \approx 2N \cdot \Phi_{\text{BM}}(E, E_c(B_0)) \cdot \Delta\theta_h^{\text{used}}. \quad (28)$$

Here Φ_{BM} must be the vertically-clipped flux integrated over $\psi \in [-w_v/(2D), +w_v/(2D)]$ (i.e. `f_psi=2` in `sync_ene()`).

When $\Delta\theta_h^{\text{slit}} \geq 2K/\gamma$ (full aperture), Approach 2 overestimates by exactly $\pi/4$, because $dP/d\theta \propto B$ and the fan-average of the local field $B(\theta) = B_0 \sqrt{1 - (\theta\gamma/K)^2}$ is $\langle B \rangle = (\pi/4)B_0$.

Hybrid implementation

The two approaches are selected automatically based on the slit-to-fan ratio:

$$\text{Use Approach 2 if } \frac{w_h}{D} < \frac{2K}{\gamma}, \quad \text{Use Approach 1 if } \frac{w_h}{D} \geq \frac{2K}{\gamma}. \quad (29)$$

Approach	E_c used	Horizontal weight	Best for
1 (arc integral)	local $E_c(s)$	$\int B(s) ds / (B\rho)$	total power
2 (peak field)	$E_c(B_0)$	$\min(w_h/D, 2K/\gamma)$	slit power

Note that the hybrid switch (Eq. 29) has a 27% discontinuity at $w_h/D = 2K/\gamma$: Approach 2 hits the boundary at $1.273\times$ exact while Approach 1 is exact. A smooth alternative is the elliptical weighting:

$$\Delta\theta_h^{\text{eff}} = \frac{K}{\gamma} \left[\arcsin u + u\sqrt{1-u^2} \right], \quad u = \min\left(1, \frac{w_h\gamma}{2DK}\right) \quad (30)$$

which $\rightarrow w_h/D$ for small slits (the rectangle is fine near the axis, since $B(\theta) \approx B_0$ there) and $\rightarrow (\pi/2)K/\gamma$ at full aperture (exact total power). The check above confirms both limits, and for the CPMU18 slit ($u=0.27$) it differs from the rectangle by only 1.2% — confirming why the small-slit results were already excellent.

3 Summary and Recommendations

- **Approach 1 (arc integration)**: integrating the BM spectral formula locally along the sinusoidal arc, weighted by the local bending angle $d\theta = |B(s)|ds/(B\rho)$, gives the exact total power $P = 633 B_0^2 L E_e^2 I [\text{W}]$ with a naturally averaged spectral shape. This is the recommended approach for **full-aperture power**.
- **Approach 2 (peak field)**: using the peak field B_0 and $E_c(B_0)$ with $\Delta\theta_h^{\text{used}} = \min(2K/\gamma, w_h/D)$ gives near-exact agreement with SRW ($\sim 2\%$) for a slit that clips the horizontal fan. This is the recommended approach for **slit-integrated power**. This is the approach to be used for the wattdog ID11 calculations, where the maximum aperture is $1.8\times 1.0 \text{ mm}^2$ at 23 m.
- A **hybrid implementation** selects automatically between the two approaches based on whether the slit is narrower or wider than $2K/\gamma$. The correction for smoothing the transition has not been implemented.
- The approximation **fails for spectral shape**: it produces a smooth BM-like continuum and misses the harmonic structure entirely. It should *not* be used for monochromatic flux, rocking-curve calculations, or any application requiring spectral detail.

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